LIDA Composting System
Software Testing Document

This document defines the testing plan, approach, environment, and structure for verifying the functionality and performance of the LIDA Composting System. It ensures that all system components including the 3D model, mobile application, and sensor simulation meet functional requirements and operate correctly when integrated.

The primary goal of this document is to:

- Establish a consistent testing framework for all team members.
- Record test cases, results, and evidence for verification and validation.
- Identify and resolve any system-level defects before final deployment.

Document Title	LIDA Composting System – Software Testing Document
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Approved By	

Test Environment

Category	Description
Hardware	Laptop (i5/i7 processor, 8 GB RAM minimum) Mobile device (Android 10+ or iOS 14+)
Software	SolidWorks 2021 / SketchUp Pro, Flutter SDK, Visual Studio Code, Android Studio
Test Data	Mock sensor data (temperature, gas, moisture) using local JSON files or MQTT broker
Network	Local Wi-Fi connection / simulated offline testing
Testing Tools	SolidWorks Motion Manager, Flutter Hot Reload, Android Emulator, Screenshot Tools
Testing Standards	IEEE 829 Software Test Documentation Standard / Agile Sprint Testing Cycle

Testing Approach

- Testing Levels:
    1. Unit Testing: Verification of individual modules (3D model components, animations, sensor logic).
    2. Integration Testing: Validation of interactions between modules (e.g., 3D models vs mobile app).
    3. System Testing: Full-system testing of performance, data flow, and UI behaviour.
- Testing Methodology:
  Manual testing and black-box approach will be used to validate outputs against requirements.
- Test Execution:
    - Each team member executes their own unit tests.
    - Thenura consolidates integration testing and system testing results.
    - Screenshots and logs will be attached as evidence.

Roles and Responsibilities

Role	Name	Responsibility
Scrum Master / Lead Tester	Thenura Kuruppuarachchi	Coordinate testing activities and compile final test document.
Animation Tester	Haritha Perera	Test motion logic and animations for paddles and condensation.
Pipeline & Sensor Tester	Senupama Deshapriya	Validate pipeline flow, drainage, and sensor data accuracy.
UI and Documentation Tester	Chrishelle Anthoneyz	Verify UI data display, app export, and prepare evidence screenshots.

Testing Tools and Techniques

- Manual Functional Testing
- Black-Box Testing
- Regression Testing (for updated models)
- Integration Simulation (3D Model vs App)
- Visual Inspection and Animation Replay

Unit Testing

Test ID	Module / Functionality	Test Objective	Input / Action	Expected Output	Actual Output	Status	Evidence (Screenshot)
UT-01	Paddle Animation	Verify bidirectional rotation	Run Motion Study	Smooth counter-rotation			
UT-02	Drainage Pipe	Check flow animation	Play animation	Moisture flows through pipe			
UT-03	Sensor Label Display	Confirm sensor labels visible	Enable label view	Gas, Temp, Moisture labels show up			
UT-04	GLB Export	Validate model export	Export to GLB format	Model opens in Flutter			

Integration Testing

Test ID	Integrated Modules	Test Objective	Expected Outcome	Actual Outcome	Status	Evidence (Screenshot)
IT-01	3D Model + App Interface	Verify GLB model loads correctly	Model renders in app without errors			
IT-02	Sensors + UI	Check live sensor updates	Sensor values update every 5 seconds			
IT-03	App + Monitor Display	Ensure app and on-model monitor sync data	Readings match on both displays			

System Testing

Test ID	Scenario / Feature	Test Objective	Expected Behaviour	Actual Behaviour	Status	Evidence (Screenshot)
ST-01	Full System Run	Verify animation, sensor, and UI run together	System runs smoothly without lag			
ST-02	App Performance	Check loading speed and UI responsiveness	App opens and renders within 5 sec			
ST-03	Data Consistency	Validate sensor data matches simulation	All values consistent with mock input			

Test Results Summary

Testing Type	Total Cases	Passed	Failed	Remarks
Unit Testing				
Integration Testing				
System Testing				

Reviewed By	Signature	Date
Supervisor (Mr. Naveed Ali)		
Scrum Master (Thenura Kuruppuarachchi)		